

Night-Time Lights

What the darkness tells us — reading Africa's development from space

Concepts, instruments, artefacts and the workflow behind today's four laboratories: from raw radiance to a validated subnational indicator.

Data Innovation For Africa (DI4A) - AfDB



SECTION 1

01

Foundations

Why should a photograph of the dark tell us anything at all?

The core idea · Definitions · The inferential chain



Light is what human activity leaks

Almost everything people do after sunset requires light — and part of that light escapes upward, into the field of view of a satellite.

That physical fact turns the night side of the planet into an observation of human presence that no census, survey or administrative register can match on all four properties at once:



Globally consistent

Same method in every country



Spatially continuous

Every ≈500 m grid cell



Politically neutral

Independent of reporting



Freely available

Open NASA / EOG products

1992

Start of the continuous digital DMSP record — three decades of history

500 m

VIIRS Day/Night Band pixel, nearly constant across the swath, data product

Nightly

Global VIIRS overpass around 01:30 local time, since 2012

For an NSO: censuses come roughly once a decade, surveys every few years — lights arrive every night, for every district.

What night-time lights are — and are not

WORKING DEFINITION



Night-time lights (NTL) are measurements of visible and near-infrared radiance travelling upward from the Earth's surface, recorded by a satellite sensor during local night, when the scene is free of direct sunlight.

What it is



A physical measurement

Radiance in $\text{nW}\cdot\text{cm}^{-2}\cdot\text{sr}^{-1}$ for calibrated sensors (VIIRS).



A consistent grid

Same method over every country — Rwanda and Nigeria alike.



A near-real-time record

Daily since 2012; monthly and annual composites, free of charge.

What it is not



A photograph

The familiar “Earth at night” is a statistical composite of many cloud-free nights.



A direct measure of people or income

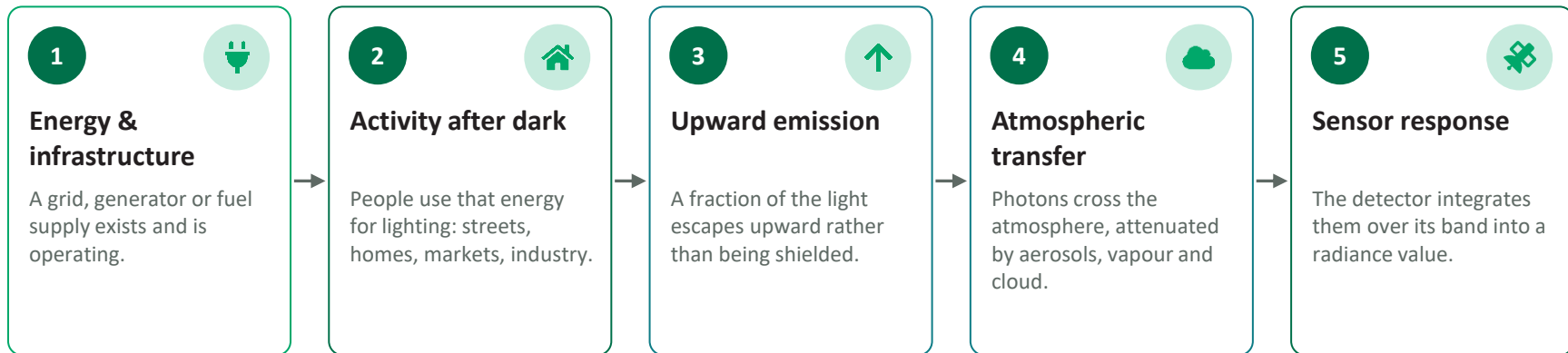
Those are inferences — each needs its own justification.



A complete record of lighting

Indoor, shielded and sideways light never reaches the sensor.

From electricity to a pixel value: five links



Where the chain breaks in African contexts — a dark pixel can mean ...

Link 1

Load-shedding or fuel shortage — people are there, power is not

Link 4

Rainy-season cloud, harmattan dust, biomass-burning smoke

Link 2–3

Solar lanterns and small home systems too faint to detect

Link 5

Rural light below the sensor's detection floor

SECTION 2

02

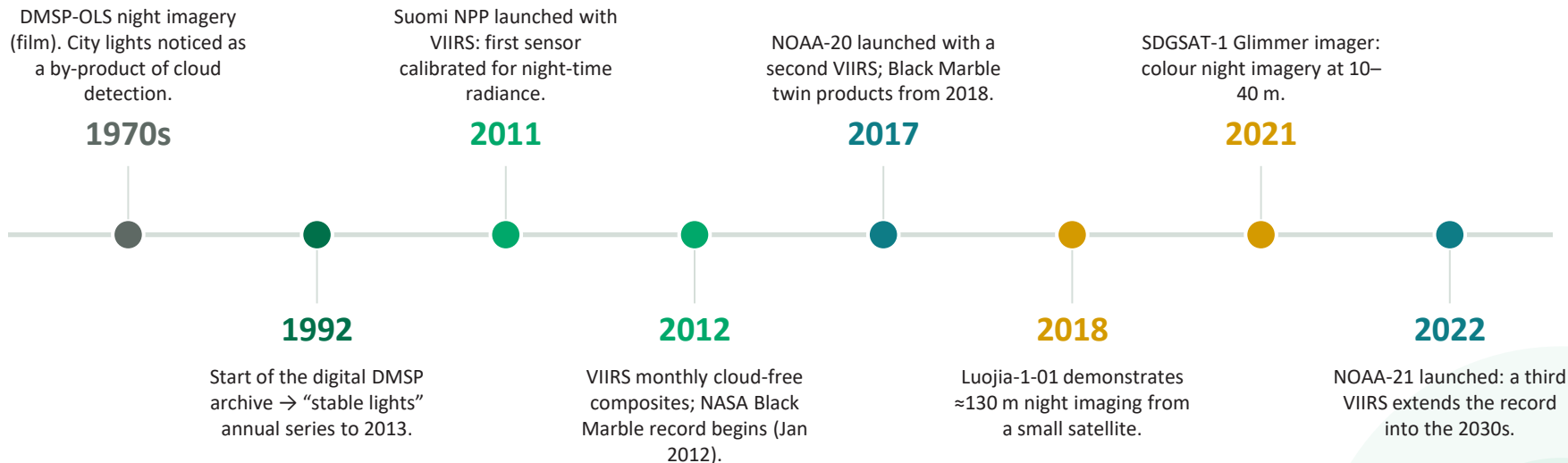
The instruments

What exactly is doing the measuring, and what can it resolve?

History · DMSP versus VIIRS · Saturation and dynamic range



Fifty years of looking down at the dark



The irony: the first instrument was built to see moonlit clouds. City lights were noise — it took two decades for anyone to treat the noise as the signal.

Two sensors, two different sciences

	DMSP-OLS · historical	VIIRS Day/Night Band · modern
Period	1992 – 2013 (archive)	2012 – present (standard)
Radiometry	6-bit digital numbers 0–63, no on-board calibration	14-bit, calibrated radiance in $\text{nW}\cdot\text{cm}^{-2}\cdot\text{sr}^{-1}$
Resolution	≈ 2.7 km, gridded at 30 arc-seconds (≈ 1 km)	500 m, nearly constant across the swath
Saturation	City cores clip at 63	Wide dynamic range, three gain stages
Sensitivity	Misses faint, rural lighting	Detects far fainter light than DMSP
Products	Annual composites only	Daily, monthly and annual — open licence



Next generation: Luojia-1-01 (≈ 130 m), SDGSAT-1 Glimmer (10 m panchromatic, 40 m colour) and commercial constellations resolve streets and distinguish lamp types by colour.

Why the brightest places say the least



DMSP returns 63 above a certain brightness — a suburb and a downtown core look identical.



VIIRS spans roughly seven orders of magnitude, with three switchable gain stages.



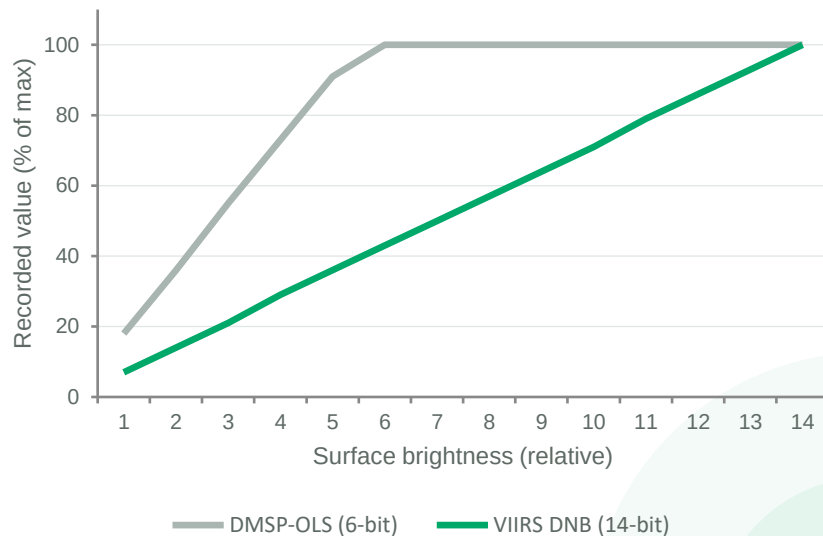
Its low noise floor makes a single lit fishing boat visible on a dark lake or ocean.



Read this before you regress

Saturation is a ceiling, not noise you can average away. The information was never recorded — correcting it means modelling what the sensor failed to see.

Sensor response as surface brightness rises (schematic)



Schematic illustration — not fitted data. VIIRS range per Elvidge et al. (2017).

SECTION 3

03

From photons to data

How does a night of observations become a number you can publish?

Processing · Products · What glows · Indicators · Example



How a night-lights image is manufactured



You will see these steps as layers at 09:30

Black Marble VNP46A2 carries the corrected radiance plus its evidence: cloud-mask flag, lunar irradiance, snow flag and a Mandatory_Quality_Flag (0 = high quality; 1–5 = poor, incl. aurora and glint).



A product, not a snapshot

Two products for the same month can legitimately differ. Always record product and version.

Sources: Román et al. (2018); NASA LAADS DAAC VNP46A2 v2 product page.

The NASA Black Marble product family

	VNP46A1 DAILY	VNP46A2 DAILY	VNP46A3 MONTHLY	VNP46A4 ANNUAL
Content	At-sensor radiance (top of atmosphere)	Moonlight-adjusted, gap-filled NTL	Composite of corrected daily NTL	Composite of corrected daily NTL
Corrections	Uncorrected — inputs & geometry only	Cloud, lunar, atmosphere, terrain, stray light	As A2, plus number of observations & quality	As A2, plus number of observations & quality
Typical use	Advanced users	Blackouts, reliability, events	Programme evaluation, seasonality	Long-run trends, subnational panels



Common to all: 15 arc-second grid (≈500 m at the equator) · 10°×10° tiles · HDF-EOS5 · from January 2012 · NOAA-20 twins named VJ146*.

Gotcha: radiance is stored as integers — multiply by the 0.1 scale factor and mask the 65535 fill value. **Alternatives:** EOG VIIRS Nighttime Lights (VNL) composites; Google Earth Engine catalogue.

What actually glows over Africa at night



Cities & settlements

SIGNAL

Streets, buildings, markets, vehicles. The dominant and most-studied part of the signal.



Corridors, ports & mines

SIGNAL

Highways, ports, airports, mining and industrial sites: the network between the bright spots.



Gas flares

MASK

Very bright, isolated points in oil & gas regions — e.g. the Niger Delta, Algeria, Libya. Mask them.



Fishing lights

CONTEXT

Lamp-attraction fishing on Lake Victoria and Lake Tanganyika; offshore fleets.



Fires

MASK

Savanna and agricultural burning: intense, short-lived and strongly seasonal.



Natural & reflected light

REMOVE

Moonlight on bright desert sand or salt pans, airglow. Not a human signal at all.

Gas flares are catalogued separately by the VIIRS Nightfire product (Earth Observation Group).

Turning an image into a number

Toy district · radiance per pixel

0.1	0.3	0.2	0.0
0.4	2.5	6.0	0.6
0.2	4.0	12.0	1.5
0.0	0.8	1.2	0.3

Shaded = above threshold $0.5 \text{ nW}\cdot\text{cm}^{-2}\cdot\text{sr}^{-1}$

30.1

Sum of Lights (SOL)

Total radiance inside a boundary. Workhorse of the economics literature; most sensitive to background noise.

8 px

Lit area

Pixels above a chosen threshold. Intuitive — but the threshold is a researcher choice.

3.6

Mean radiance (lit)

Brightness independent of extent. Separates a district getting brighter from one getting bigger.



Raise the threshold to 1.0 → lit area falls from 8 to 6 pixels (–25%). Always report product, version, boundary, threshold and masks — and test the threshold.

Illustrative values. Light concentration (Gini of radiance) is a fourth family, used for spatial inequality.

One question, end to end



Did the 2023 mini-grid programme actually bring light to its target villages?

1 State the claim in light terms

Electrified villages should show more nights above their background radiance after commissioning.

2 Choose the product

VNP46A3 monthly to date the change; VNP46A2 daily if you also want reliability. Record the version.

3 Define the units

Settlement footprints plus a small buffer — not whole districts, where a few lit villages vanish.

4 Pick indicator & control

Mean radiance and share of lit nights; controls = similar villages outside the programme.

5 Read the difference

A step change in programme villages that is absent in controls is evidence; a common rise is regional.

6 Try to break it

Check fires, flares, moon and season; compare with utility connection or programme monitoring records.

SECTION 4

04

Applications

Which statistical questions can this data actually answer?

SDG map · Economy · Energy access · Urbanisation · Crises



Where lights can support SDG monitoring



SDG 1

Poverty & wealth mapping

Small-area estimates combining lights with survey data



SDG 7

Electricity access & reliability

Locate unserved settlements; track how often lights are on



SDG 8

Economic activity

Subnational GDP downscaling; short-term activity signals



SDG 9

Infrastructure

Corridors, ports, industrial and special economic zones



SDG 11

Urbanisation

Urban extent, sprawl versus densification, new towns



SDG 16 · humanitarian

Conflict & disasters

Blackouts, displacement signals, recovery tracking



A complement, not a replacement. Lights add spatial detail, frequency and independent verification to official statistics — they do not replace the census or the survey.

Measuring the economy from orbit

What the evidence shows



Henderson, Storeygard & Weil (2012)

Lights carry real information on economic growth and can improve GDP estimates where national accounts are weak.



Bruederle & Hodler (2018)

Lights track household wealth, education and health at the local level, using DHS survey clusters in Africa.



Martinez (2022)

Lights used as an independent benchmark to test the credibility of reported GDP growth.

Why the lights–output link is not stable



Sector mix: agriculture and mining create value with little light



Income level: bright cores saturate or plateau



Lighting technology: LED retrofits change light per watt



Power supply: load-shedding dims lights, not output

WHERE LIGHTS HELP MOST

Subnational

High-frequency

Informal economy

Sources: Henderson et al. (2012) AER 102(2); Bruederle & Hodler (2018) PLOS ONE 13(9); Martinez (2022) JPE 130(10).

Energy access and SDG 7: Africa's defining gap

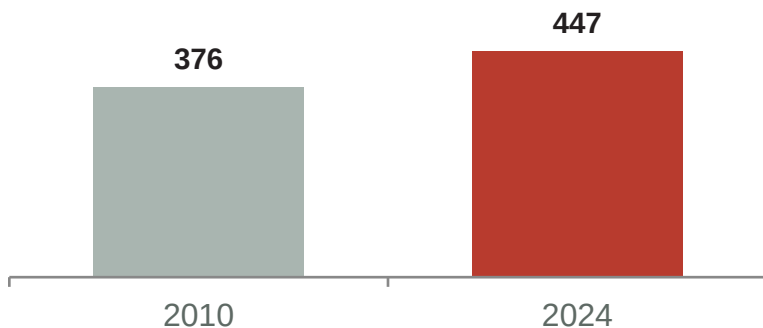
655 M

people without electricity
worldwide (2024)

> 560 M

of them in sub-Saharan Africa

Rural sub-Saharan Africa without electricity (millions)



Locate the unserved

Settlements visible by day but dark at night are candidate unelectrified communities.



Verify programmes

A grid extension that produces no new light has not, in a meaningful sense, been delivered.



Measure reliability

Daily data reveal how often lights are on — load-shedding a binary “connected” statistic hides.



Plan the last mile

Distance to lit corridors informs grid extension vs mini-grid vs stand-alone solar.

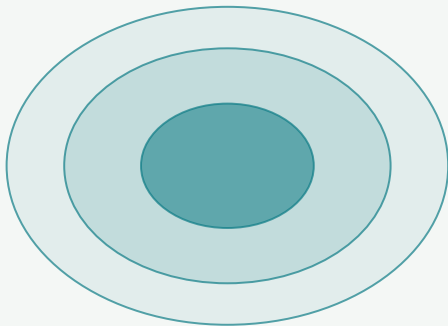
Mission 300 (World Bank Group & AfDB): 300 M people connected by 2030 — monitoring at that scale needs spatial evidence.

Sources: Tracking SDG 7: The Energy Progress Report (IEA, IRENA, UN DESA, World Bank, WHO), 24 June 2026; HREA, Min & O’Keeffe (Univ. of Michigan).

Urbanisation and the shape of a city

Sprawl

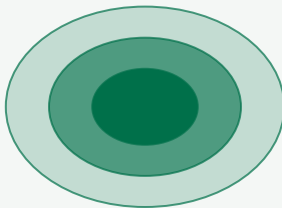
Footprint expands outward at low intensity



Lit area +40% · Radiance +6%

Densification

The same footprint intensifies



Lit area +6% · Radiance +40%

Same Sum of Lights — two very different cities (illustrative).



Map the footprint where registers are missing

Lights reveal built-up extent where cadastral records or recent maps do not exist.



Separate sprawl from densification

Compare growth in lit area with growth in radiance — it matters for transport and services.



Detect new towns early

Mining boomtowns, economic zones and displacement settlements appear before official geography.



Pair with population grids

Use WorldPop or GHSL — bright pixels are not people.

When the lights go out: conflict and disaster



Conflict · Sudan, 2023

After fighting began in Khartoum in April 2023, the capital's lights fell to the noise level by May (EOG). UNOSAT mapped power outages from NTL in June 2023.



Conflict · Syria, 2011–2014

Li & Li (2014) measured a loss of more than 70% of night-time light nationally, with losses tracking displacement province by province.



Disaster · Puerto Rico, 2017

After Hurricane Maria, NASA Black Marble daily data mapped the blackout and tracked restoration neighbourhood by neighbourhood.



The interpretive rule: absence of light is not absence of people

Darkness may mean displacement, destruction, fuel shortage, curfew, a grid failure — or a decision to switch off street lights. The satellite cannot tell which. Corroboration with ground information is not optional, and uncertainty must be stated.

Sources: Payne Institute / EOG (Sept 2024); UNOSAT (June 2023); Li & Li (2014) IJRS 35(18); Román et al. (2019) on Puerto Rico.

SECTION 5

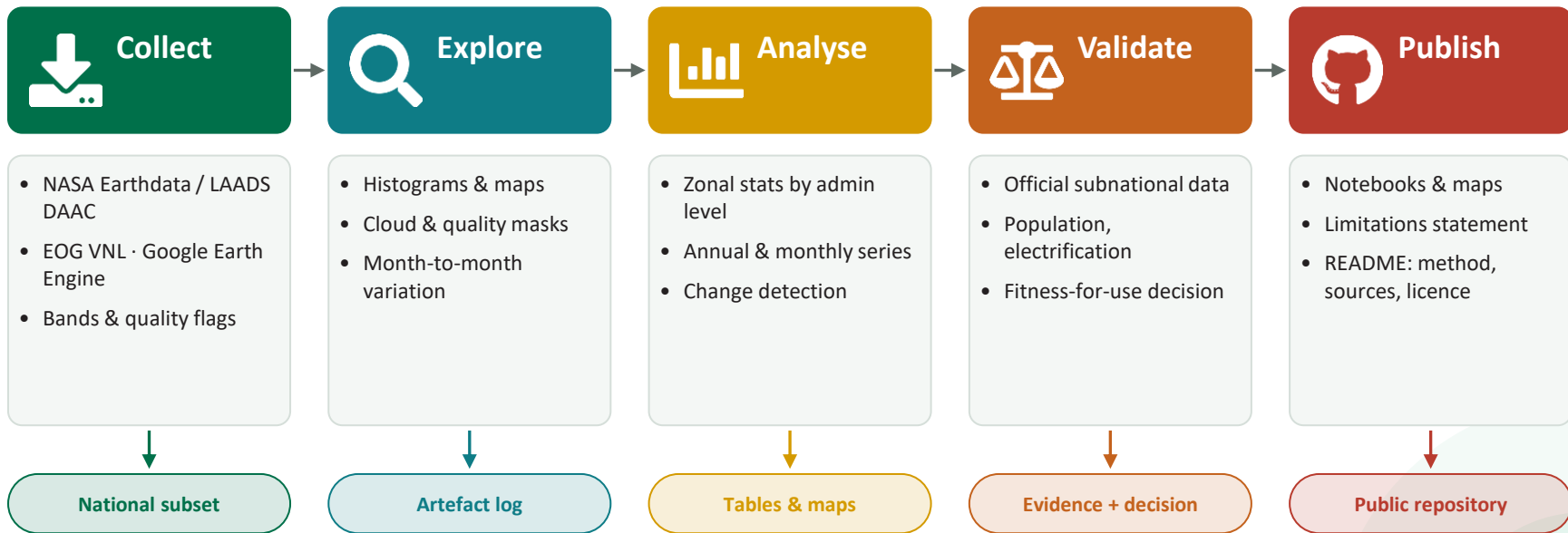
05

From talk to lab

Workflow · Choosing a product · Validation · Checklist · Discussion



Five stages, one reproducible pipeline



Example open-source stack: *rasterio / xarray (read), geopandas (boundaries), rasterstats or exactextract (zonal statistics), matplotlib (maps).*

Which cadence for which question?



Annual

VNP46A4 · EOG VNL annual

STRENGTH

Cleanest series; noise and ephemeral lights largely filtered.

WATCH OUT FOR

A change cannot be dated within the year.

TYPICAL NSO USE

Subnational panels, long-run trends, urban expansion, poverty mapping.



Monthly

VNP46A3 · EOG VNL monthly

STRENGTH

Dates a change to within a season.

WATCH OUT FOR

Rainy-season months may rest on very few valid nights.

TYPICAL NSO USE

Programme evaluation, seasonal activity, conflict & recovery timelines.



Daily

VNP46A2

STRENGTH

Every night, corrected for moon, atmosphere and terrain.

WATCH OUT FOR

Large volumes; a single night is noisy — aggregate first.

TYPICAL NSO USE

Blackouts, disaster response, reliability of electricity supply.



Rule of thumb: start annual. The finer the cadence, the more corrections you must trust and the more noise you must handle.

A checklist for defensible analysis

1

Name product and version

VNP46A4 v2, VNL V2.x — they are not interchangeable.

2

Never splice sensors naively

Any DMSP–VIIRS series needs documented harmonisation.

3

Report and test the threshold

Show the finding survives a reasonable range.

4

Mask what you are not studying

Background noise, gas flares, fishing lights, fires.

5

Normalise deliberately

Per capita, per km², or neither — state and defend it.

6

Validate against the ground

Census, surveys, utility records — independent sources.

7

Prefer changes to levels

Differences cancel much of the calibration error.

8

Publish code and limitations

README with method, sources, licence and caveats.

**THE GOLDEN RULE**

State, in one sentence, what would have to be true about lighting behaviour for your conclusion to hold.

If you cannot write that sentence, you do not yet have a result.



Share it: a documented, reusable repository is how your office contributes to the STG17 methodological guideline on new data sources.

Five things to take into the lab

1

Night lights are a proxy, never the quantity of interest — name the assumption that links them to your question.

2

The instrument defines the science: DMSP saturates, VIIRS calibrates, and neither sees blue light.

3

The data is most valuable where official statistics are thinnest: subnational, high-frequency, in crises and informal economies.

4

Darkness is not emptiness — off-grid light, cloud and load-shedding all look alike from orbit.

5

Validate against independent sources, and publish your limitations with your results.



The darkness tells us a great deal. It does not tell us everything — and it never speaks unprompted.

The vocabulary you will meet today



Radiance

Energy leaving a surface per unit area and solid angle; for NTL in $\text{nW}\cdot\text{cm}^{-2}\cdot\text{sr}^{-1}$.



Composite

Grid assembled from many cloud-free nights — not a photograph of one night.



Sum of Lights (SOL)

Total radiance summed over all pixels inside a boundary.



BRDF correction

Adjustment for how reflected (moon)light varies with viewing and lighting angles.



Nadir

Point directly beneath the satellite, where the pixel footprint is smallest.



Digital number (DN)

Uncalibrated sensor count with no physical unit; DMSP reports 0–63.



Stable lights

Product with fires, aurora and other unstable sources removed.



Blooming / overflow

Apparent spread of light beyond its source (scattering, geolocation error).



Zonal statistics

Summary of raster values (sum, mean, count) within each polygon.



Skyglow

Diffuse brightening of the night sky by artificial light scattered in the atmosphere.

Selected references and data sources

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DATA & TOOLS



NASA Black Marble

LAADS DAAC · Earthdata



Earth Observation Group

VNL composites, Nightfire



Google Earth Engine

Hosted, analysis-ready archive



HREA

Settlement electricity access



World Bank Open Nighttime Lights

Tutorials & notebooks